HANOI UNIVERSITY OF
SCIENCE AND TECHNOLOGY

SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

DATA VISUALIZATION REPORT

Telecom Customer Churn Analysis

Dataset, Exploratory Analysis, and Visualization Technique Plan

Course: Data Visualization

Class: 168068

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Group repository: https://github.com/NguyenNgocMinh30012005/DataVis_2025.2

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Abstract

This report defines the analytical and visualization structure for a telecom customer churn analysis project. The dataset was downloaded from Kaggle and then split into churned and non-churned customer files for analysis. The current draft follows the course capstone guideline by documenting the dataset source, exploratory analysis, feature grouping, churn-rate interpretation, Random Forest prediction workflow, and the visualization techniques applied to the final dashboard screenshots. Some recommended techniques remain future improvements, but the included screenshots reflect the group's final reporting artifact.

Contents

1 Introduction

1.1 Business Context

Customer churn is a critical issue for subscription-based telecom companies. When customers leave, the business loses recurring revenue and may need to spend additional resources to acquire replacement customers. A churn analysis report should therefore help analysts and managers identify high-risk customer groups and understand which customer attributes are associated with a higher churn rate.

1.2 Project Objectives

The project has four objectives:

1. Analyze customer churn patterns across customer, account, service, and experience dimensions.
2. Compare churned and non-churned customers through exploratory data analysis.
3. Build interpretable feature groups that support churn-rate comparison.
4. Explain how course visualization techniques are applied to the final dashboard pages and summarize retention recommendations based on the observed signals.

1.3 Current Scope

At this stage, this report is prepared as an Overleaf-compatible final report draft. It contains the cover page, table of contents, dataset source, dataset statistics, EDA plan, model workflow, expected insight narrative, and a technique application section aligned with the course guideline in `guideline.md`. The final dashboard screenshots are included from the group's final reporting artifact in the `img/` folder. Not every recommended technique could be implemented before submission, so the report distinguishes implemented design choices from remaining future improvements.

2 Dataset

2.1 Dataset Source

The dataset is a large customer churn prediction dataset with simulated, privacy-preserving customer records. It is available on Kaggle and can be accessed at: https://www.kaggle.com/datasets/isandeep06/customer-churn-prediction-dataset-1m/data. It can also be searched through Google Dataset Search: https://datasetsearch.research.google.com/search?query=Customer%20Churn%20Prediction%20Dataset%201M.

The source-level statistics are summarized in Table 1.

Table 1: Kaggle source listing summary

Item	Value
Dataset title	Customer Churn Prediction Dataset 1M
Source platform	Kaggle
Original scale	1,000,000 customer records
Original feature count	28 features + 1 target variable
Main target	Customer churn label

Table 2 summarizes all variables included in the dataset, grouped into their respective categories along with a brief description of each feature.

Table 2: Dataset feature description

Category	Column	Description
Customer Demographics	customer_id	Unique customer identifier
	signup_date	Customer registration date
	age	Customer age (18–90)
	gender	Gender: Male, Female, Other
	annual_income	Annual income in USD
	education	Education level
	marital_status	Marital status
	dependents	Number of dependents (0–5)
	senior_citizen	1 if age ≥ 65, else 0
Account Information	tenure	Months with company (1–72)
	contract	Contract type: month_to_month, one_year, two_year
	payment_method	Payment method
	paperless_billing	Paperless billing: Yes/No
Service Usage & Billing	monthlycharges	Monthly service charges ($20–$200)
	totalcharges	Total charges to date
	num_services	Number of subscribed services (1–6)
	has_phone_service	1 if has phone service
	has_internet_service	1 if has internet service
	has_online_security	1 if has online security
	has_online_backup	1 if has online backup
	has_device_protection	1 if has device protection
	has_tech_support	1 if has tech support
	has_streaming_tv	1 if has streaming TV
	has_streaming_movies	1 if has streaming movies
Customer Behavior & Risk	customer_satisfaction	Satisfaction score (1–10)
	num_complaints	Complaints in last year (0–8)
	num_service_calls	Service calls last month (0–12)
	late_payments	Late payments last 3 months (0–5)
	avg_monthly_gb	Average monthly data usage (GB)
	days_since_last_interaction	Days since last contact (1–365)
	credit_score	Credit score (300–850)
Target Variable	churn	Target: 1 if churned, 0 if retained

2.2 Preprocessing

The dataset was cleaned by removing records with missing values and split into churn and non-churn subsets for the group analysis.

The resulting datasets' sizes are as shown in Table 3.

2.3 Working Dataset Descriptive Statistics

After cleaning and splitting the downloaded data, the working dataset used in this report contains 840,765 rows and 32 columns, with no missing values. Numeric descriptive statistics are shown in Table 4.

Table 3: Current dataset size

Segment	Number of customers
Churn	83,362
Not churn	757,403
Total	840,765

Table 4: Numeric statistics in the working dataset

Feature	Mean	Median	Min	Max
Age	44.72	44.00	18.00	90.00
Annual income	58,809.80	48,973.70	20,000.00	250,000.00
Dependents	0.80	1.00	0.00	5.00
Tenure	22.40	16.00	1.00	72.00
Senior citizen	0.20	0.00	0.00	1.00
Monthly charges	86.43	85.46	20.00	854.96
Total charges	1,838.74	1,250.23	16.30	16,252.86
Num services	2.56	2.00	1.00	6.00
Has phone service	0.77	1.00	0.00	1.00
Has internet service	0.85	1.00	0.00	1.00
Has online security	0.34	0.00	0.00	1.00
Has online backup	0.43	0.00	0.00	1.00
Has device protection	0.30	0.00	0.00	1.00
Has tech support	0.50	0.00	0.00	1.00
Has streaming TV	0.60	1.00	0.00	1.00
Has streaming movies	0.55	1.00	0.00	1.00
Customer satisfaction	6.16	7.00	1.00	9.00
Number of complaints	0.70	1.00	0.00	7.00
Number of service calls	1.76	1.00	0.00	12.00
Late payments	0.40	0.00	0.00	5.00
Average monthly GB	39.08	27.75	0.00	557.82
Days since last interaction	44.47	31.00	1.00	365.00
Credit score	678.54	680.00	300.00	850.00
Churn	0.10	0.00	0.00	1.00

Additionally, table 3 shows that the baseline churn rate is approximately 9.92%. This number is used as the reference point when interpreting group-level churn rates. A group above this baseline should be considered higher risk, while a group below the baseline is relatively lower risk.

3 Feature Selection and Grouping

3.1 Rationale

The dataset contains a diverse set of demographic, account, service usage, and customer behavior variables that may influence customer churn. To facilitate a structured exploratory data analysis, features are organized into logical categories based on their business context. This categorization improves the interpretability of the analysis by allowing related variables to be examined together and helps identify patterns associated with customer retention and churn. In addition, selected continuous variables are grouped into meaningful ranges where appropriate to simplify visualization and discussion of customer characteristics. The same feature organization is maintained throughout the project to ensure consistency between the exploratory analysis and the predictive modeling stages.

3.2 Selected Overall Model Features

The overall churn prediction model uses the full dataset with both churned and non-churned customers. The selected raw features are:

- tenure
- contract
- customer_satisfaction
- num_complaints
- num_service_calls
- late_payments
- monthlycharges
- num_services
- has_tech_support

These variables cover four main analytical dimensions: customer tenure, contract commitment, customer experience, and service/payment behavior.

3.3 Correlation Matrix Screening

Before finalizing the Phase 2 feature set, the encoded overall dataset was screened with a Pearson correlation matrix. The full matrix is exported by the pipeline as `overall_encoded_correlation_matrix.csv`. Because the full matrix is too wide for the report, Table 5 summarizes the strongest encoded-feature correlations with the churn target using shortened feature labels. These values should be read as univariate screening signals, not as final causal evidence.

Table 5: Top encoded features by absolute correlation with churn

Feature signal	**Corr.**	**|Corr.|**	**Interpretation**
`contract: two-year`	-0.123686	0.123686	Two-year contracts are associated with lower churn.
`contract: one-year`	0.100979	0.100979	One-year contracts have higher churn than the two-year group.
`satisfaction: 1-3`	0.080981	0.080981	Low satisfaction is associated with higher churn.
`contract: month-to-month`	0.078836	0.078836	Month-to-month contracts are associated with higher churn.
`complaints: <= 1`	-0.066236	0.066236	Fewer complaints are associated with lower churn.
`complaints: > 1`	0.066236	0.066236	More complaints are associated with higher churn.
`satisfaction: 7-9`	-0.063678	0.063678	High satisfaction is associated with lower churn.
`service calls: <= 3`	-0.060627	0.060627	Fewer service calls are associated with lower churn.
`service calls: > 3`	0.060627	0.060627	More service calls are associated with higher churn.
`has tech support`	-0.045187	0.045187	Technical support is associated with lower churn.

At the feature-family level, the largest absolute correlations are concentrated in contract type, customer satisfaction, complaints, service calls, technical support, and late payments. Tenure and monthly charges are much weaker in this screening. This supports using the customer-experience and contract-related features as the primary explanatory features in the dashboard, while treating low-correlation variables as supporting context rather than dominant churn drivers.

Table 6: Feature-family correlation summary

Feature family	Max |corr.|	Mean |corr.|	Top encoded feature
contract	0.123686	0.101167	contract_two_year
customer_satisfaction_group	0.080981	0.049728	satisfaction_1_3
num_complaints_group	0.066236	0.066236	complaints_le_1
num_service_calls_group	0.060627	0.060627	service_calls_le_3
has_tech_support	0.045187	0.045187	has_tech_support
late_payments_group	0.041126	0.033550	late_0
num_services_group	0.027798	0.021552	services_1_2
tenure_group	0.011551	0.005833	tenure_32_plus
monthlycharges_group	0.000050	0.000050	monthly_ge_200

The churn-only dataset is not used for target-correlation screening because it contains only churned customers. In that file, the churn target has no class variation, so correlation with churn is undefined. Churn-only correlations are therefore useful only for profiling relationships inside the churned-customer group.

3.4 Selected Churn-Only Support Features

The churn-only file contains only customers who already churned, so it is not used to train a standalone binary classifier. Instead, it supports profiling and explanation of the churned customer group. The selected support features are:

- annual_income
- education
- contract
- paperless_billing
- num_complaints
- num_service_calls
- late_payments
- monthlycharges

3.5 Feature Grouping Rules

Table 7: Raw features and grouping rules used by the project pipeline

Raw feature	Grouped feature	Grouping rule
annual_income	annual_income_group	$<$ 30K, 30–60K, 60–90K, $>$ 90K
education	education_group	master/PhD = graduate; others = non-graduate
tenure	tenure_group	1–6, 7–15, 16–31, 32+ months
contract	contract	original contract category, one-hot encoded for modeling
paperless_billing	paperless_billing_flag	No = 0; Yes = 1
customer_satisfaction	customer_satisfaction_group	1–3, 4–6, 7–9
num_complaints	num_complaints_group	≤ 1, > 1
num_service_calls	num_service_calls_group	≤ 3, > 3
late_payments	late_payments_group	0, 1–2, 3+
monthlycharges	monthlycharges_group	< 200, ≥ 200

Raw feature	Grouped feature	Grouping rule
num_services	num_services_group	1–2, 3–4, 5–6
has_tech_support	has_tech_support	binary 0/1 feature

3.6 Interpretability Benefit

The selected features are not only model inputs; they also provide a structure for explaining churn. For example, satisfaction and complaints describe customer experience, contract and tenure describe commitment, and service/payment features describe operational behavior. This organization makes the final analysis easier to communicate to non-technical audiences.

4 Prediction Workflow

4.1 Random Forest Pipeline

The pipeline uses the cleaned churn dataset, creates grouped features, encodes categorical variables, trains a Random Forest classifier, and exports model outputs for later analysis.

4.2 Model Evaluation Summary

Table 8 reports the Random Forest test metrics:

Table 8: Random Forest test metrics from the group repository

Metric	Value
ROC AUC	0.676510
Accuracy	0.761694
Precision	0.187745
Recall	0.421945
F1-score	0.259863

4.3 Model Performance Interpretation

The model performance should be interpreted in the context of churn prediction, rather than as a generic classification score. In the working dataset, the number of churned customers is 83,362 out of 840,765 customers, which gives a baseline churn rate of approximately 9.92%. This means that the target class is strongly imbalanced: only about 10 out of every 100 customers are churned customers.

For this reason, accuracy is not the primary evaluation metric. A naive model that predicts every customer as non-churn would already obtain about 90.08% accuracy, but it would detect no churned customers. The Random Forest accuracy of 76.17% should therefore not be interpreted as the main indicator of model quality. More informative metrics are precision, recall, F1-score, ROC AUC, and precision-recall based metrics. In this run, the recall of 42.19% means that the model identifies about 42% of the actual churned customers, while the low precision reflects the difficulty of separating a relatively small churn class from a much larger non-churn class.

The model is also constrained by the Phase 2 feature design. The initial model uses interpretable, dashboard-friendly grouped features such as contract type, customer satisfaction group, complaint group, service-call group, late-payment group, monthly-charge group, tenure group, service-count group, and technical support. These variables contain useful directional signals, but they do not capture all possible churn drivers. For example, the dataset does not include detailed behavioral histories such as plan changes, discount history, competitor offers, support ticket text, customer sentiment, or detailed payment timelines.

Another limitation is that several numeric variables are grouped into bins before modeling. This helps the dashboard remain readable and supports clear segment-level interpretation, but it can remove continuous detail that a predictive model might otherwise use. Therefore, the Random Forest model should be treated as a decision-support component for prioritizing retention actions, not as a perfect churn prediction system.

The selected threshold is chosen on the validation set to improve the balance between precision and recall. Lowering the threshold would identify more churned customers but would also increase false positives. Raising the threshold would improve precision but miss more churned customers. This trade-off is acceptable for a retention dashboard, where the objective is to highlight customers or segments that deserve earlier attention from customer support or retention teams.

Future model improvement should focus on using raw numeric features together with grouped features, testing gradient boosting models such as XGBoost, LightGBM, or CatBoost, tuning hyperparameters against F1 or PR-AUC rather than accuracy, handling class imbalance with class weights or resampling inside the training pipeline, and adding richer behavioral features if they become available.

5 Exploratory Data Analysis Plan

5.1 How Churn-Rate Charts Should Be Read

In churn-rate charts, a bar does not represent a distribution that sums to 100% across groups. Each bar represents the churn rate inside one group:

$$\text{Churn Rate}_{group} = \frac{\text{Number of churned customers in the group}}{\text{Total customers in the group}} \times 100\%.$$

This means each group should be compared independently against the baseline churn rate of approximately 9.92%.

5.2 Customer Experience Features

Table 9: Planned EDA interpretation by feature

Feature	Preliminary interpretation
Customer satisfaction	Higher satisfaction tends to lean toward non-churn. Low satisfaction groups are expected to have higher churn risk.
Number of complaints	More complaints are associated with a higher churn rate and should be treated as a strong customer-experience risk signal.
Number of service calls	More service calls may indicate unresolved service problems and higher churn risk.
Late payments	More late payments are associated with higher churn rate, although the signal is weaker than satisfaction and complaints.
Average monthly GB	Usage groups have relatively similar churn rates; this feature is weak as a standalone churn driver.
Days since last interaction	Churn rates are expected to be nearly flat across time-since-interaction groups, so this feature should be combined with other signals.

Feature	Preliminary interpretation
Credit score	Credit-score groups show small churn-rate differences and are not expected to be a major standalone churn signal.

5.3 Planned EDA Visuals

For each selected feature, the final report should include two complementary visuals:

1. Customer count by churn status for each group.
2. Churn rate for each group.

This structure separates customer distribution from churn risk. A large segment does not automatically mean high churn risk, and a small segment can still be important if its churn rate is much higher than the baseline.

5.4 Priority of Feature Interpretation

The feature interpretation should be ordered by actionability. Customer experience and payment-behavior variables are more useful for retention planning because they can trigger operational actions. Usage and credit variables should be treated as supporting context unless they show a clear above-baseline churn-rate pattern.

1. Customer satisfaction.
2. Number of complaints.
3. Number of service calls.
4. Late payments.
5. Average monthly GB.
6. Days since last interaction.
7. Credit score.

6 Visualization Design Principles for Churn Analysis

6.1 Audience and Analytical Questions

The visualization artifact is designed for analysts and customer-retention stakeholders. The intended use is not only to report how many customers churned, but to identify which customer groups are above the baseline churn rate and which groups should be prioritized for retention actions.

The core analytical questions are:

1. What is the overall churn baseline?
2. Which customer segments are large enough to matter operationally?
3. Which segments have churn rates above the baseline?
4. Which features are actionable for a retention team?
5. How do model outputs support, but not replace, business interpretation?

6.2 Metric Semantics

The report separates distribution metrics from risk metrics. This distinction is essential because a segment can contain many customers but still have an average churn rate, while a smaller segment can have a much higher churn rate.

Table 10: Metric definitions used in the visualization design

Metric	Definition	Interpretation rule
Customer count	Number of customers in a segment.	Measures segment size, not churn risk.
Churn count	Number of churned customers in a segment.	Useful for workload sizing, but affected by segment size.
Churn rate	Churn count divided by all customers in the same segment.	Measures within-segment risk and should be compared with the 9.92% baseline.
Share of churned customers	Churned customers in a segment divided by all churned customers.	Shows composition of churned customers only; values can sum to 100%.
Predicted churn probability	Model-estimated probability of churn for customers or groups.	Useful for prioritization, not as a final business decision by itself.
Risk level	Grouped model or rule-based risk category.	Should use consistent thresholds and colors across pages.

6.3 Chart Selection by Analytical Task

The chart type should follow the task, not the visual style preference. The current churn project mainly contains table data, so the most relevant chart families are bar charts, distribution charts, relationship views, and summary tables.

Table 11: Recommended chart types for churn analysis tasks

Task	Recommended chart	Churn example
Compare counts	Grouped bar chart	Churn vs. non-churn by satisfaction group
Compare risk	Bar chart with baseline	Churn rate by complaint group
Show distribution	Histogram or box plot	Credit score or monthly charges
Show composition	100% stacked bar	Share of churned customers by segment
Show relationship	Scatter, heatmap, or small multiples	Satisfaction vs. complaints
Show model behavior	Metric cards and confusion matrix	AUC, precision, recall, F1-score

6.4 Visual Encoding Rules

The design should use encodings that are easy to decode:

- Use position and bar length for comparing counts, churn rates, and feature importance.
- Use color hue for categorical distinction such as churn versus non-churn or low/medium/high risk.

- Use a baseline line for churn-rate charts so above-baseline groups can be detected quickly.
- Avoid pie charts, donut charts, and 3D charts when accurate comparison is required.
- Avoid using color as the only carrier of meaning; pair color with labels, ordering, or reference lines.

6.5 Color System

The color system should encode meaning consistently. A single default blue for all charts is visually clean but does not communicate risk hierarchy. The recommended semantic palette is:

Table 12: Suggested semantic color palette

Meaning	Color role	Suggested hex
Churn / high risk	Highlight and warning	#D55E00
Medium risk	Intermediate priority	#E69F00
Low risk	Safer segment	#009E73
Neutral count / selected bar	Main neutral blue	#2563EB
Not churn / inactive context	Supporting gray-blue	#64748B
Background	Page background	#F8FAFC
Gridline / border	Low-contrast structure	#E5E7EB

These colors should remain stable across pages. For example, high-risk customers should not be blue on one page and orange on another page. If color is used to highlight a high-risk segment, the chart should also include a label or baseline marker so that the message is still readable without color.

6.6 Page-Level Visualization Structure

The final visualization artifact can be organized as a compact story rather than a collection of independent charts:

1. **Overview**: show total customers, churn count, non-churn count, baseline churn rate, and risk-level distribution.
2. **Churn drivers**: compare feature-group churn rates against the baseline and show top model feature importance.
3. **Churn-only composition**: show the share of churned customers by selected segment, while clearly stating that the denominator is churned customers only.
4. **Model evaluation**: show AUC, accuracy, precision, recall, F1-score, confusion matrix, and a short interpretation of model limitations.
5. **Prediction and retention priority**: show predicted churn customers, risk levels, and the features that define priority groups.

6.7 Final Dashboard Screenshots

The final reporting artifact contains three dashboard pages. The pages do not implement every recommended technique from the design checklist, but they are used as the final version for the project report due to time constraints. The most important requirement is that the interpretation remains clear: KPI cards summarize the problem size, bar charts compare segment risk, and model outputs support retention prioritization.

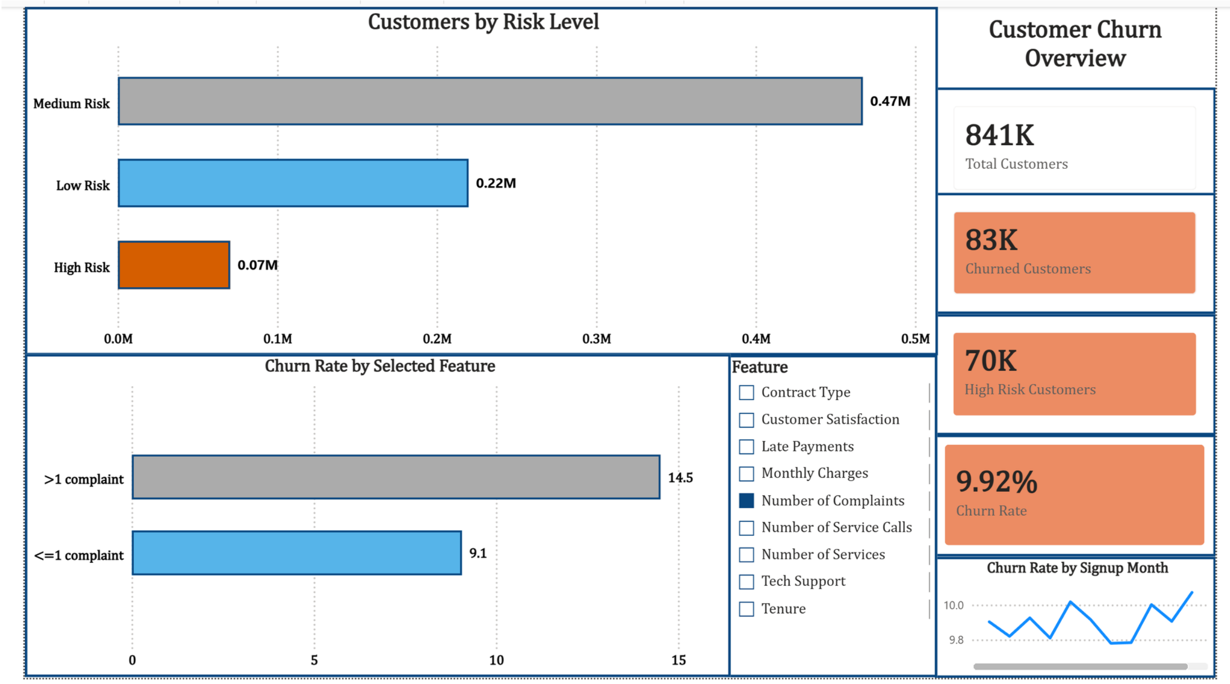

Figure 1: Final dashboard page 1: customer churn overview, KPI cards, risk-level distribution, selected-feature churn rate, and signup-month trend.

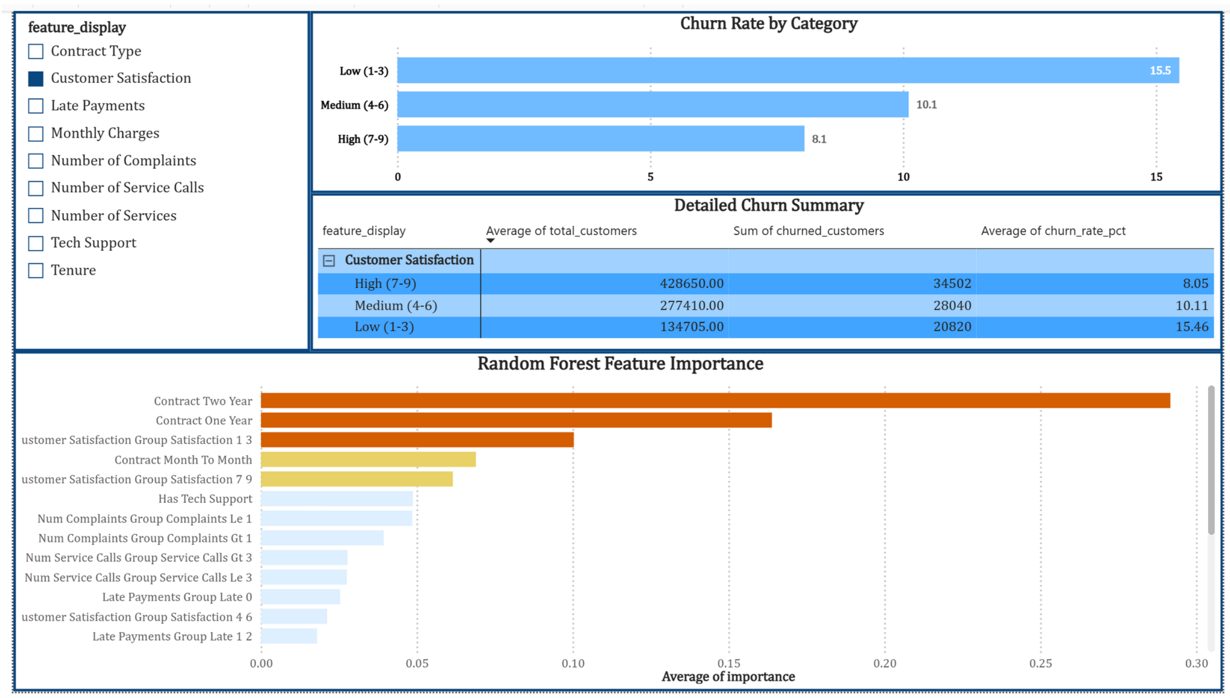

Figure 2: Final dashboard page 2: churn driver detail page with feature filter, churn-rate-by-category view, detailed summary table, and Random Forest feature importance.

6.8 Figure Quality Checklist

When reading the final screenshots, each chart should be checked against the following criteria:

- The title states either the analytical question or the main insight.
- The denominator of every percentage or rate is explicit.
- Count views and churn-rate views are not mixed into one ambiguous metric.
- Churn-rate charts include the 9.92% baseline reference.
- Category order is meaningful: ordinal groups keep natural order, while nominal groups are sorted by value only when ranking is intended.
- Colors are semantic and consistent across pages.
- The chart avoids 3D effects, unnecessary decoration, and hard-to-read legends.

7 Technique Application According to Course Guideline

7.1 Guideline Alignment

The course guideline at `https://datalab.vn/articles/2021-09/IT5425-quan-tri-du-lieu-va-truc-quan-hoa` requires the report to explain which visualization techniques from the course are applied to each relevant chart or page. It also requires the dataset to be searchable through Google Dataset Search, which is already documented in the dataset source section. This section follows that structure using the final dashboard screenshots prepared by the group for reporting.

The technique mapping below uses the course guideline, the local course slides, and the distilled note file `Slide data viz/2024/notes/00_distilled_knowledge.md`. The most relevant lectures are introduction to visualization, visual models and visual encoding, graphical perception, table-data visualization, figure design, and storytelling with data.

- Introduction to data visualization.
- Visual model and visual encoding.
- Graphical perception.
- Visualization for table data.
- Principles of figure design.
- Storytelling with data.

7.2 Reusable Technique-Application Template

For each final chart or page, the report documents the design decision in the following format:

- **Chart/Page**: name of the final page or visual.
- **Technique / principle applied**: related course concept, such as visual encoding, common baseline, proportional ink, or storytelling.
- **How applied**: which variable is mapped to which mark, axis, color, or interaction.
- **Notes / adjustments**: limitations, denominator, or reason for not using another chart type.

Example: in a churn-rate-by-satisfaction chart, satisfaction groups should be placed in ordinal order, churn rate should be encoded by bar length on a common axis, and a 9.92% baseline line should identify above-baseline groups. The notes should state that these bars are within-group churn rates and therefore are not expected to sum to 100%.

7.3 Chapter 1: Overview of Data Visualization

Techniques / principles applied. The project uses visualization for both exploratory and explanatory purposes. Exploratory charts help compare churned and non-churned customers across feature groups. Explanatory charts communicate which groups are above or below the baseline churn rate.

How applied in the dashboard. The final overview page starts with customer and churn KPIs, risk-level distribution, selected-feature churn-rate comparison, and a signup-month trend. This allows the audience to distinguish "how many customers are in a group" from "how risky that group is".

Notes / adjustments. The dataset is large and customer-level, so the dashboard should avoid showing raw customer rows as the main view. Aggregated segment views are more suitable for decision making.

7.4 Chapter 2: Visual Models and Visual Encoding

Techniques / principles applied. The selected variables cover several measurement levels: nominal variables such as contract type and payment method, ordinal or discrete variables such as customer satisfaction and complaint counts, and quantitative variables such as monthly charges, average monthly GB, credit score, and days since last interaction. The dashboard charts therefore use marks and channels that match each variable type.

How applied in the dashboard. Bar marks should encode grouped categories. Position on a common axis and bar length should encode counts and churn rates. Color hue should separate churn from non-churn when both categories are shown, while a distinct highlight color can mark groups whose churn rate is above the 9.92% baseline.

Notes / adjustments. The report should avoid over-encoding. A chart should not use color, shape, and size for the same distinction unless the redundancy improves accessibility.

7.5 Chapter 3: Graphical Perception

Techniques / principles applied. The design should prefer encodings that are easier to compare, especially position and length. Preattentive color can guide attention to high-risk groups. Gestalt principles such as proximity and similarity should group related charts and filters.

How applied in the dashboard. For the EDA pages, customer-count bars and churn-rate bars should be placed near each other for the same feature. Bars should share a consistent baseline so that users can compare magnitudes accurately. Above-baseline churn groups should be highlighted consistently across all feature pages.

Notes / adjustments. Color must not be the only signal for risk. Labels, baseline lines, and consistent ordering should also indicate which groups are high risk.

7.6 Chapter 4: Visualization for Table and Multidimensional Data

Techniques / principles applied. The project mainly uses table-data visualization: amounts, distributions, proportions, and relationships between grouped features and churn. The current EDA plan separates customer distribution from churn risk.

How applied in the dashboard. Counts by churn status should use grouped bars. Churn rates by feature group should use independent bar charts with the baseline churn rate as a reference line. Numeric variables can use grouped bins, histograms, or box plots before being summarized into business-readable groups. Relationships among multiple features can be explored with filtered views or small multiples.

Notes / adjustments. Churn-rate bars across groups do not need to sum to 100%. Each bar represents the churn rate inside that group, not the share of total churn. Stacked bars should only be used when the parts form a meaningful whole.

7.7 Chapter 5 and Chapter 7: Graph and Map Visualization

Techniques / principles applied. These chapters are not central to the current dataset because the data does not contain explicit network edges or geographic locations.

How applied in the dashboard. No graph layout or map view is used in the current version.

Notes / adjustments. If the group later adds geographic or relationship data, this section should be expanded with map encoding or graph-layout choices.

7.8 Chapter 6: Principles of Figure Design

Techniques / principles applied. The final figures are evaluated against proportional ink, clear titles and captions, color accessibility, multi-panel consistency, and a balance between data and context. Unnecessary 3D charts should be avoided.

How applied in the dashboard. Bars should start from zero when encoding counts or rates. Each chart should include a title that states the analytical question, a y-axis label for the metric, and a data-source note where appropriate. Multi-panel pages should keep the same color mapping for churn and non-churn. Risk highlights should use a colorblind-aware palette and direct labels when possible.

Notes / adjustments. The dataset is aggregated for visualization, so overlapping points are less of a problem than in scatter plots. If scatter plots are added later, transparency, binning, or sampling should be used to reduce overlap.

7.9 Chapter 8: Interactive Visualization

Techniques / principles applied. The relevant interaction techniques are filtering, selection, and view changes. Animation is not required because the available dataset is not time-series focused.

How applied in the dashboard. Users should be able to filter by contract type, tenure group, satisfaction group, complaint group, service-call group, and payment behavior. Selecting a segment should update both the distribution view and the churn-rate view so that the effect of the filter is visible.

Notes / adjustments. Interactions should support comparison, not hide the baseline. The 9.92% overall churn-rate reference should remain visible whenever a filtered churn rate is shown.

7.10 Chapter 9: Storytelling with Data

Techniques / principles applied. The report should move from context to evidence and then to action. The target audience is a business or customer-retention team that needs to identify which customer groups deserve attention.

How applied in the dashboard. The recommended narrative order is: introduce the churn problem, show the baseline churn rate, compare churned and non-churned customers, identify the strongest customer-experience signals, connect EDA with model outputs, and end with retention recommendations.

Notes / adjustments. Each page should have a clear message title. The final version should replace generic chart titles such as "Churn by Feature" with insight-oriented titles such as "Low satisfaction customers churn above the baseline" when the numbers support that statement.

8 Expected Insights and Recommendations

8.1 High-Priority Signals

The strongest expected signals are related to customer experience:

- Low customer satisfaction.
- High number of complaints.
- High number of service calls.

These features are directly actionable because they can guide the customer support and retention teams toward customers who may be experiencing service problems.

8.2 Moderate Signal

Late payments are expected to have a moderate positive relationship with churn. This feature should be interpreted together with satisfaction, complaint history, and service calls before it is used for retention targeting.

8.3 Weak Standalone Signals

Average monthly GB usage, days since last interaction, and credit score are expected to have weak standalone separation between churn and non-churn customers. They may still be useful as filters or as supporting variables in a multivariate model.

8.4 Prediction Workflow Insight

The Random Forest workflow in the group repository supports the EDA findings by using interpretable churn-related features such as contract, satisfaction, complaints, service calls, late payments, monthly charges, service count, and tech support. Because the churn class is imbalanced, the model is more suitable for ranking retention priority than for making a final yes/no business decision without human review.

8.5 Retention Recommendations

1. Prioritize customers with low satisfaction scores for proactive support.
2. Create a complaint escalation view for customers with multiple complaints.
3. Monitor repeated service calls as a sign of unresolved technical issues.
4. Combine late-payment signals with experience-related signals before launching retention campaigns.
5. Avoid drawing strong conclusions from usage volume or credit score alone.

8.6 Storytelling Flow for the Final Report

The final report narrative should move from context to action:

1. Establish why churn matters for a subscription-based telecom company.
2. State the baseline churn rate and total customer population.
3. Separate segment size from within-segment churn risk.
4. Highlight above-baseline customer groups using EDA evidence.
5. Connect the EDA findings with model evaluation and feature importance.
6. End with retention actions that are tied to customer experience variables.

This narrative avoids treating the model as the only result. The model supports prioritization, while the visualization explains which segments are actionable and why they deserve attention.

9 Limitations and Next Steps

9.1 Current Limitations

- Current interpretations are based mainly on single-feature EDA.
- The model metrics should be interpreted carefully because the churn class is imbalanced; accuracy is less informative than recall, precision, F1-score, ROC AUC, and PR-AUC for this task.
- Several numeric variables are grouped for dashboard readability, which improves interpretation but can reduce predictive detail for the model.
- Some likely churn drivers are not available in the dataset, including detailed payment history, plan-change events, discount exposure, competitor offers, and support-ticket content.
- The final dashboard screenshots are included, but not every recommended course technique could be implemented before submission.
- Some visual polish remains possible, especially consistent typography, semantic color use, and clearer page-level titles.

9.2 Next Steps

1. Re-check all churn-rate values against the latest cleaned dataset.
2. Add more multivariate analysis after the single-feature EDA is stable.
3. Review Random Forest feature importance and high-risk customer examples.
4. Test raw numeric features together with grouped features and compare Random Forest with XGBoost, LightGBM, or CatBoost.
5. Add PR-AUC, threshold analysis, and class-imbalance handling to the modeling evaluation if the predictive workflow is extended.
6. If more time is available, refine chart titles, color semantics, and churn-rate denominators directly in the dashboard artifact.
7. Add more final insight paragraphs after the group completes the oral presentation narrative.

10 Conclusion

This draft establishes the structure for the final telecom churn analysis report. The analysis should focus on churn rate by interpretable customer groups and clearly distinguish distribution from risk. The visualization design should follow the course guideline by connecting each chart to a relevant principle from visual encoding, graphical perception, figure design, table-data visualization, interaction, or storytelling. Based on the current EDA direction, customer experience variables are expected to be more informative than usage or credit-score variables when analyzed individually. The group repository also provides a Random Forest workflow that can be used to rank customers by retention priority.

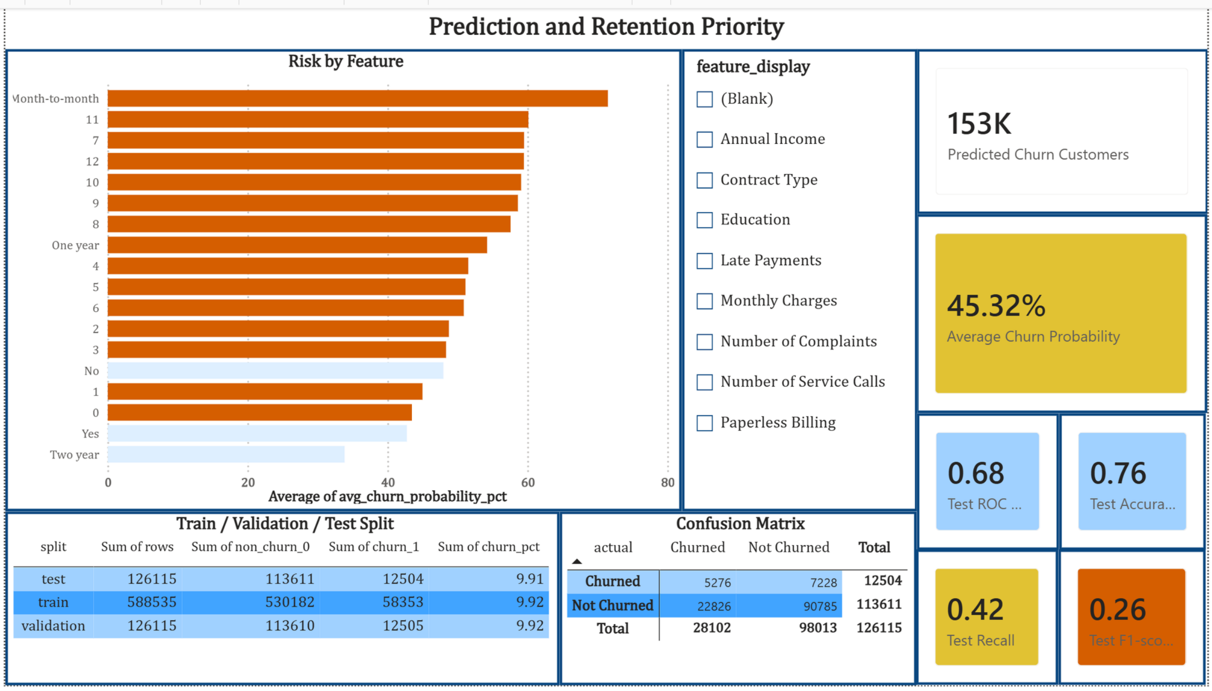

Figure 3: Final dashboard page 3: prediction and retention priority page with risk-by-feature view, predicted churn KPIs, train/validation/test split, and confusion matrix.